

Math723 – Differential Equations I

Fall 2016

Final Exam

You must show all of your work to get a credit for a correct answer.

1. Let $u(x, y)$ be a harmonic function in $\mathbb{R}^2$ and suppose that

$$\int_{\mathbb{R}^2} |\nabla u|^2 \, dxdy < \infty$$

Prove that u is a constant function.

2. Let u be a positive harmonic function in the unit ball $B(\mathbf{0}, 1)$ in $\mathbb{R}^n$. Show that

$$|D(\ln(u))| \leq M, \quad \text{in } B(\mathbf{0}, 1/2),$$

for some constant M depending only on the dimension n .

3. Show that for the solution of the heat equation

$$\begin{cases} u_t - u_{xx} = 0, & \text{in } \mathbb{R} \times (0, \infty), \\ u(x, 0) = f(x), & \text{in } \mathbb{R}, \end{cases}$$

where f is square integrable, there exists a constant α independent of f and t such that for all $t > 0$

$$\max_x |u_x(x, t)| \leq \alpha t^{-\frac{3}{4}} \left(\int_{\mathbb{R}} |f(x)|^2 \, dx \right)^{1/2}.$$

4. Find the solution of the initial value problem

$$\begin{cases} u_{tt} - u_{xx} + u = 0, & \text{in } \mathbb{R} \times (0, \infty), \\ u(x, 0) = f(x), & \text{in } \mathbb{R}, \\ u_t(x, 0) = g(x), & \text{in } \mathbb{R}, \end{cases}$$

where $f, g \in C_0^\infty(\mathbb{R})$.

5. Solve using the characteristics

$$\begin{cases} u_t - x_2 u_{x_1} + x_1 u_{x_2} = 0, & \text{in } \mathbb{R}^2 \times (0, \infty), \\ u(x_1, x_2, 0) = e^{-x_2^2}, & \text{in } \mathbb{R}^2. \end{cases}$$